



ISO/TC 12, Quantities and units

Secretariat: SIS

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Report of the 17th Meeting of ISO/TC 12 held on 2020-05-28 on Zoom

1. Opening of the meeting (2020-05-28, 14:00 h CEST)

The Chair Prof. Pendrill welcomed the participants and opened the meeting.

The meeting was held on Zoom in lieu of the face-face meeting originally scheduled for Saint Petersburg, Russian Federation on the same date due to the Covid-19 restrictions.

2. Roll call of delegates

19 delegates attended this meeting via Zoom, representing **8 P-Members**: China, Czech Republic, Finland, Germany, Portugal, Russian Federation, Sweden, and United-States; and **two Liaisons**: CIE and IEC/TC 25. The **ISO/CS Technical Programme Manager**, Ms. Ferrés was present in the meeting. See N 1369, Attendance list.

3. Presentation on the Code of Conduct

Direct link to [Code of Conduct](#)

ISO/TC 12 reviewed the ISO Code of Conduct and agreed to abide by its principles in all of its work. The presentation had been circulated to the committee in document N 1344.

4. Adoption of the agenda

Doc. ISO/TC 12 N 1356

The draft agenda was adopted as presented. A number of additional documents were noted; all had been circulated recently and were flagged up for review under the relevant agenda items during the course of the meeting:

Document N 1366 to Agenda item 8, *Report of ISO/TC 12 /WG 21 – General*;

Document N 1364 to agenda item 9, *Report of IEC/TC 25/JWG 2 – Revision and amendment of IEC related parts of ISO/IEC 80000 series linked to ISO/TC 12*;

Document N 1357, N 1358, N 1360, N 1361, N 1362 to Agenda item 11, *Reports from Liaisons*.

Reports had been received from the following committees and organisations;

ISO/TC 43/SC 1 - Noise

ISO/TC 69 – *Applications of statistical methods*

IEC/TC 25 – *Quantities and units*

CIE – *International Commission on Illumination*

ICRU – *International Commission on Radiation Units and Measurements*

Document N 1359 to Agenda item 13.5, *IEC/PWI 80000-17, Quantities and units – Part 17: Time dependency (IEC lead)*

Document N 1363, *Input to IEC-ISO SM 2 TF, Joint Smart Manufacturing Standards Map Task Force*

5. Appointment of the drafting committee

Mr. Elster (Sweden) and Mr. Hübner (Germany) were appointed to assist the secretariat in the drafting of the resolutions.

Resolution 130, Zoom 1/2020

**ISO/TC 12, Quantities and units****6. Report of the Chair and of the Secretariat**

Doc. ISO/TC 12 N 1365 (Chair's report) N 1341 (report of the secretariat)

The Chair, Prof. Pendrill, presented doc N 1365, *Report from Chair to the plenary meeting*. He pointed out that, much of the work of ISO/TC 12 with the ISO 80000-series (particularly part 1) in the past year had benefited from extensive mutual contacts and exchanges between TC12 and two other international bodies active in the field: the JCGM (responsible for VIM, Vocabulary of International Metrology, and the CCU (responsible for the SI brochure).

He gave an overview of recent trends in science and technology regarding quantities and units. One such sector is the health care sector where ISO/TC 12 has an active collaboration with ISO/TC 215, *Health informatics*. Dr. Krystek commented on the slide regarding *Quantities & units in e-health* and remarked that IEC/TC 25 no longer participates in the work in this project.

The secretary presented the report of the secretariat, N 1341.

7. Business Plan for ISO/TC 12

Doc. ISO/TC 12 N 1353

This 9th edition of the Business Plan had been updated according to comments from Finland presented in doc. N 1282.

Dr. Knopf (Germany) proposed that the first paragraph should be changed to read:

"In society, processes and products will acquire better interoperability, improved communication and can be more reliably traded, and ensured safe when tested with measurement expressed in terms of harmonised quantities, units and symbols. Quantities and mathematical operations, the particular focus of ISO/TC 12, are key concepts of all exact sciences and related applications, and constitute the language of physics, chemistry, engineering and mathematics and other sciences, having impact far beyond the commercial and industrial dimensions."
(additions underlined).

ISO/TC 12 resolved to send the above text for an 8-week Committee Internal Ballot for approval.

Resolution 131, Zoom 2/2020

8. Report of ISO/TC 12 /WG 21 – General

Doc. ISO/TC 12 N 1253

The secretary of WG 21, Mr. Radhe, reported on the activities of the working group since the last ISO/TC 12 meeting and the decisions from the meeting held the day before, doc N 1366.

ISO/TC 12 reviewed the proposals and approved them. The proposals were to send out Committee Internal Ballots to receive consensus on:

- to delete the definitions from ISO/IEC Guide 99 (VIM) in ISO 80000-1;
- to replace the text in clause 6 of ISO 80000-1 by a reference to the SI Brochure, thus making the SI brochure a normative reference in this;
- to replace the numerical values in Annex A with a general reference to CODATA;
- to collect writing and printing rules in ISO 80000-1.

ISO/TC 12 decided to task the secretariat to prepare a new CD for ISO 80000-1 after the CIBs had closed.

Resolution 132, Zoom 3/2020

Resolution 133, Zoom 4/2020

Resolution 134, Zoom 5/2020

Resolution 135, Zoom 6/2020

Resolution 136, Zoom 7/2020

ISO/TC 12, Quantities and units**9. Report of IEC/TC 25/JWG 2 – Revision and amendment of IEC-related parts of ISO/IEC 80000 series linked to ISO/TC 12**

Doc. ISO/TC 12 N 1364

Prof. Obdržálek, convenor of the JWG 2, reported on the activities of JWG 2. He added that IEC 80000-15, *Logarithmic and related quantities, and their units* is currently out for comments in JWG 2, closing 2020-6-20. ISO/TC 12 thanked Prof. Obdržálek for his report.

10. Information on IEC Coordination group for measurement science

Doc. ISO/TC 12 N 1346, N 1367

ISO/TPM, Ms. Ferrés, informed on the structure of JCGM, Joint Committee for Guides in Metrology, see doc. N 1367.

The Chair reported on the IEC Coordination group for measurement science, see doc. N 1346 where he participates as invited guest.

11. Reports from liaisons

Doc. ISO/TC 12 N 1357, N 1358, N 1360, N 1361, N 1362

ISO/TC 12 had received liaison reports from the following committees:

ISO/TC 43/SC 1 – *Noise*, N 1360ISO/TC 69 – *Applications of statistical methods*, N 1357IEC/TC 25 – *Quantities and units*, N 1361CIE – *International Commission on Illumination*, N 1358ICRU – *International Commission on Radiation Units and Measurements*, 1362**12. Status of all items of the programme of work and action to be taken**

ISO/CS TPM Ms. Ferrés explained the time schedule for ISO 80000-1. DIS ballot to be initiated on 2020-10-21, publication 2022-10-22.

13. Follow-up on work

Doc. ISO/TC 12 N 1312, N 1337, N 1349, N 1354, N 1355, N 1359, N 1363

13.1 ISO/CD 80000-1, Quantities and units – Part 1: General

See Agenda Item 8.

13.2 IEC/DIS 80000-6, Quantities and units – Part 6: Electromagnetism (IEC lead)

See Agenda item 9.

13.3 IEC/CD 80000-15, Quantities and units – Part 15: Logarithmic and related quantities, and their units (IEC lead)

See Agenda item 9.

13.4 IEC/PWI 80000-16, Quantities and units – Part 16: Printing and writing rules (IEC lead)

See Agenda items 8 and 9.

13.5 IEC/PWI 80000-17, Quantities and units – Part 17: Time dependency (IEC lead)

See Agenda item 9.

ISO/TC 12, Quantities and units**13.6 Other issues arising during 80000 revision**

The secretariat has received comments at the FDIS stage and by standard users on the published standards listed below.

ISO 80000-2 *Quantities and units — Part 2: Mathematics*: Euler number etc.

ISO 80000-3 *Quantities and units — Part 3: Space and time*: Time and duration etc

ISO 80000-9, *Quantities and units — Part 9: Physical chemistry and molecular physics*: Formula

ISO 80000-11 *Quantities and units — Part 11: Characteristic numbers*: Formulas and explanations

ISO 80000-12, *Quantities and units — Part 12: Condensed matter physics*

ISO/CS TPM Ms. Ferrés explained the procedures for correcting standards, see doc N 1367, last slide. The secretary had compiled doc. N 1354, but this document needs to be further edited.

ISO/TC 12 decided to form a task force to review the comments under the leadership of Prof. Pendrill (Sweden). The members of the Task Force are Dr. Ehrlich (USA), Prof. Grimvall (Sweden), Dr. Knopf (Germany), Prof. Obdržálek (Czech Republic) and Dr. Pellegrino (Portugal). The task force will report to ISO/TC 12 by 2021-03-31.

Resolution 137, Zoom 8/2020

13.7 Input for revision of ISO/IEC Directives, Part 2: Principles and rules for the structuring of ISO and IEC documents

The secretary reported that the ISO/IEC Directives *Part 2: Principles and rules for the structuring of ISO and IEC documents*, is under revision. The current edition was published 2016. Annex B deals with quantities and units and there are also other rules from the ISO and IEC 80000 series in the main body of the text. See doc. N 1355 for comments submitted to ISO/CS on the Directives Part 2.

ISO/TC 12 decided to form a task force to deliver comments and proposals for updates to ISO/IEC Directives Part 2. Mr. Radhe (Sweden) will be the convenor and the members of the Task Force are Prof. Obdržálek (Czech Republic) and Dr. Pellegrino (Portugal). The group will report to ISO/CS by 2020-05-31.

Resolution 138, Zoom 9/2020

13.8 Input to IEC-ISO SM 2 TF, Joint Smart Manufacturing Standards Map Task Force

The secretary reported that the proposal for the ISO/TC 12 contribution to IEC/ISO SM2 TF as presented in doc. N 1349 had been approved. This document also includes the standards that IEC/TC 25 had reported. The document submitted to IEC/ISO SM2 TF only included the ISO/TC 12 standards.

14. Work items on which no progress is being made – Status and action to be taken

All items had made progress since the last meeting.

15. Items for future work

There were no discussions under this agenda item.

16. Requirements concerning a subsequent meeting

ISO/TC 12 accepted the kind invitation from GOST to host the next meeting on 2021-05-25 and 2021-05-26 in Saint Petersburg, the Russian Federation.

Resolution 139, Zoom 10/2020

17. Any other business

No other business was discussed.

18. Approval of resolutions

The resolutions were read out loud and approved.



ISO/TC 12, *Quantities and units*

19. Closure of the meeting (16:30 CEST)

The Chair thanked the delegates for their active participation and closed the meeting.